Docker networking

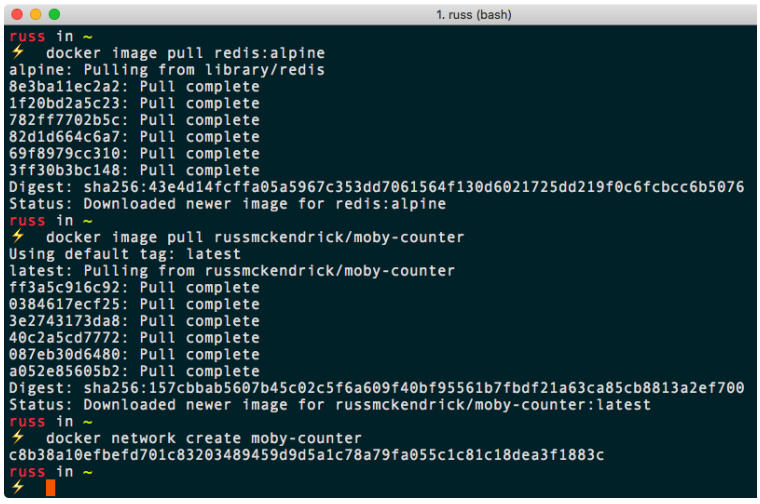
So far, we have been launching our containers on a single flat shared network. Although we have not talked about it yet, this means the containers we have been launching would have been able to communicate with each other without having to use any of the host networking.

Rather than going into detail now, let's work through an example. We are going to be running a two-container application; the first container will be running Redis, and the second, our application, which uses the Redis container to store a system state.

Redis *is an in-memory data structure store that can be used as a database, cache, or message broker. It supports different levels of on-disk persistence.*
Before we launch our application, let's download the container images we will be using, and also create the network:

```
$ docker image pull redis:alpine
$ docker image pull russmckendrick/moby-counter
$ docker network create moby-counter
```

You should see something similar to the following Terminal output:



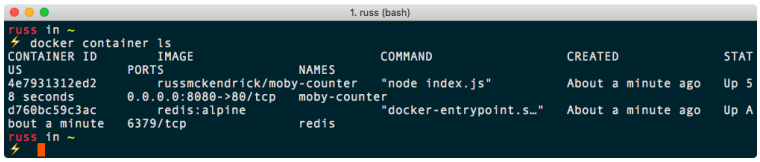
Now that we have our images pulled and our network created, we can launch our containers, starting with the Redis one:

```
$ docker container run -d --name redis --network moby-counter redis:alpine
```

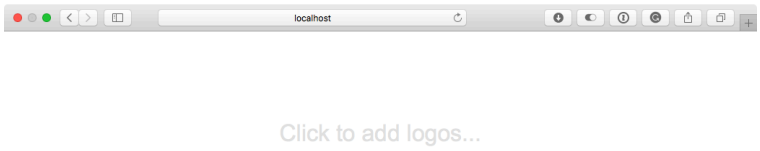
As you can see, we used the `--network` flag to define the network that our container was launched in. Now that the Redis container is launched, we can launch the application container by running the following:

```
$ docker container run -d --name moby-counter --network moby-counter -p 8080:80 rus
```

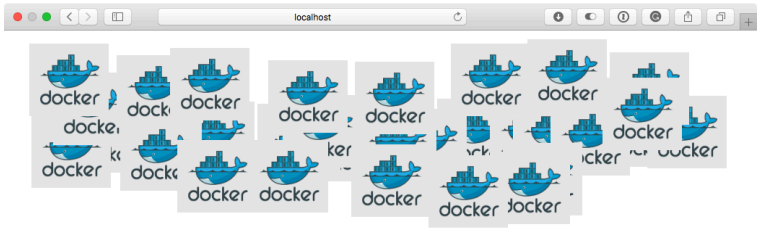
Again, we launched the container into the `moby-counter` network; this time, we mapped port `8080` to port `80` on the container. Note that we did not need to worry about exposing any ports of the Redis container. That is because the Redis image comes with some defaults that expose the default port, which is `6379` for us. This can be seen by running `docker container ls`:



All that remains now is to access the application; to do this, open your browser and go to `http://localhost:8080/`. You should be greeted by a mostly blank page, with the message **Click to add logos**:



Clicking anywhere on the page will add Docker logos, so click away:



So what is happening? The application that is being served from the moby-counter container is making a connection to the `redis` container, and using the service to store the on-screen coordinates of each of the logos that you place on the screen by clicking.

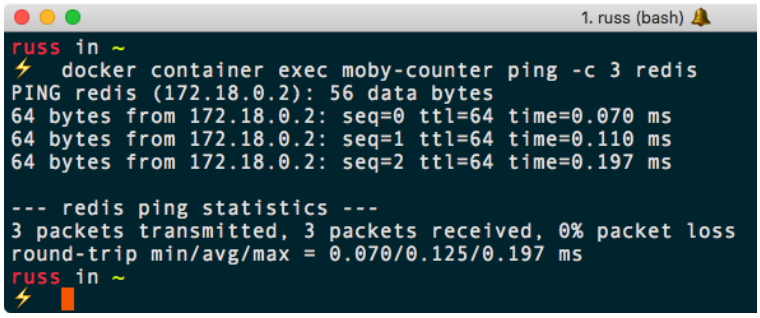
How is the moby-counter application connecting to the `redis` container? Well, in the `server.js` file, the following default values are being set:

```
var port = opts.redis_port || process.env.USE_REDIS_PORT || 6379
var host = opts.redis_host || process.env.USE_REDIS_HOST || 'redis'
```

This means that the `moby-counter` application is looking to connect to a host called `redis` on port `6379`. Let's try using the exec command to ping the `redis` container from the `moby-counter` application and see what we get:

```
$ docker container exec moby-counter ping -c 3 redis
```

You should see something similar to the following output:



As you can see, the `moby-counter` container resolves `redis` to the IP address of the `redis` container, which is `172.18.0.2`. You may be thinking that the application's host file contains an entry for the `redis` container; let's take a look using the following command:

```
$ docker container exec moby-counter cat /etc/hosts
```

This returns the contents of `/etc/hosts`, which, in my case, looks like the following:

```
127.0.0.1 localhost
::1 localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.18.0.3 4e7931312ed2
```

Other than the entry at the end, which is actually the IP address resolving to the hostname of the local container, `4e7931312ed2` is the ID of the container; there is no sign of an entry for `redis`. Next, let's check `/etc/resolv.conf` by running the following:

```
$ docker container exec moby-counter cat /etc/resolv.conf
```

This returns what we are looking for; as you can see, we are using a local `nameserver`:

```
nameserver 127.0.0.11
options ndots:0
```

Let's perform a DNS lookup on `redis` against `127.0.0.11` using the following command:

```
$ docker container exec moby-counter nslookup redis 127.0.0.11
```

This returns the IP address of the `redis` container:

```
Server: 127.0.0.11
Address 1: 127.0.0.11

Name: redis
Address 1: 172.18.0.2 redis.moby-counter
```

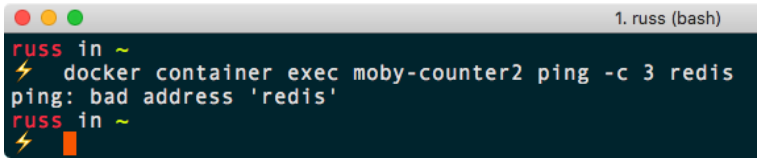
Let's create a second network and launch another application container:

```
$ docker network create moby-counter2
$ docker run -itd --name moby-counter2 --network moby-counter2 -p 9090:80 russmcken
```

Now that we have the second application container up and running, let's try pinging the `redis` container from it:

```
$ docker container exec moby-counter2 ping -c 3 redis
```

In my case, I get the following error:



Let's check the `resolv.conf` file to see if the same nameserver is being used already, as follows:

```
$ docker container exec moby-counter2 cat /etc/resolv.conf
```

As you can see from the following output, the nameserver is indeed in use already:

```
nameserver 127.0.0.11
options ndots:0
```

As we have launched the `moby-counter2` container in a different network to that where the container named `redis` is running, we cannot resolve the hostname of the container, so it returns a bad address error:

```
$ docker container exec moby-counter2 nslookup redis 127.0.0.11
Server: 127.0.0.11
Address 1: 127.0.0.11

nslookup: can't resolve 'redis': Name does not resolve
```

Let's look at launching a second Redis server in our second network; as we have already discussed, we cannot have two containers with the same name, so let's creatively name it `redis2`.

As our application is configured to connect to a container that resolves to `redis`, does this mean we will have to make changes to our application container? No, but Docker has you covered.

While you cannot have two containers with the same names, as we have already discovered, our second network is running completely isolated from our first network, meaning that we can still use the DNS name of `redis`. To do this, we need to add the `-- network-alias` flag as follows:

```
$ docker container run -d --name redis2 --network moby-counter2 --network-alias red
```

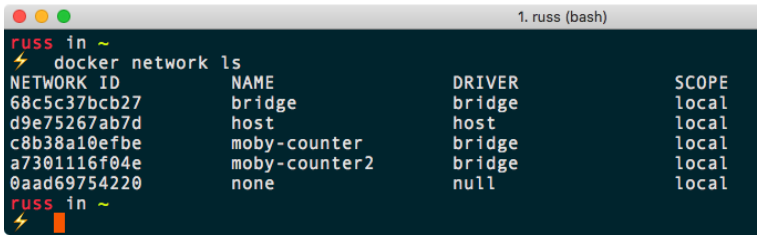
As you can see, we have named the container `redis2`, but set the `--network-alias` to be `redis`; this means that when we perform the lookup, we see the correct IP address returned:

```
$ docker container exec moby-counter2 nslookup redis 127.0.0.1
Server: 127.0.0.1
Address 1: 127.0.0.1 localhost

Name: redis
Address 1: 172.19.0.3 redis2.moby-counter2
```

As you can see, `redis` is actually an alias for `redis2.moby-counter2`, which then resolves to `172.19.0.3`.

Now we should have two applications running side by side in their own isolated networks on your local Docker host, accessible at `http://localhost:8080/` and `http://localhost:9090/`. Running `docker network ls` will display all of the networks configured on your Docker host, including the default networks:



You can find out more information about the configuration of the networks by running the following `inspect` command:

```
$ docker network inspect moby-counter
```

Running the preceding command returns the following JSON array:

```
[
    {
        "Name": "moby-counter",
        "Id": "c8b38a10efbefd701c83203489459d9d5a1c78a79fa055c1c81c18dea3f1883c",
        "Created": "2018-08-26T11:51:09.7958001Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
```

```
        "Containers": {
            "4e7931312ed299ed9132f3553e0518db79b4c36c43d36e88306aed7f6f9749d8": {
                "Name": "moby-counter",
                "EndpointID": "dc83770ae0939c98416ee69d939b30a1da391b11d14012c8188b
                "MacAddress": "02:42:ac:12:00:03",
                "IPv4Address": "172.18.0.3/16",
                "IPv6Address": ""
            },
            "d760bc59c3ac5f9ba8b7aa8e9f61fd21ce0b8982f3a85db888a5bcf103bedf6e": {
                "Name": "redis",
                "EndpointID": "5af2bfd1ce486e38a9c5cddf9e16878fdb91389cc122cfef62d5
                "MacAddress": "02:42:ac:12:00:02",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
            }
        },
        "Options": {},
        "Labels": {}
    }
]
```

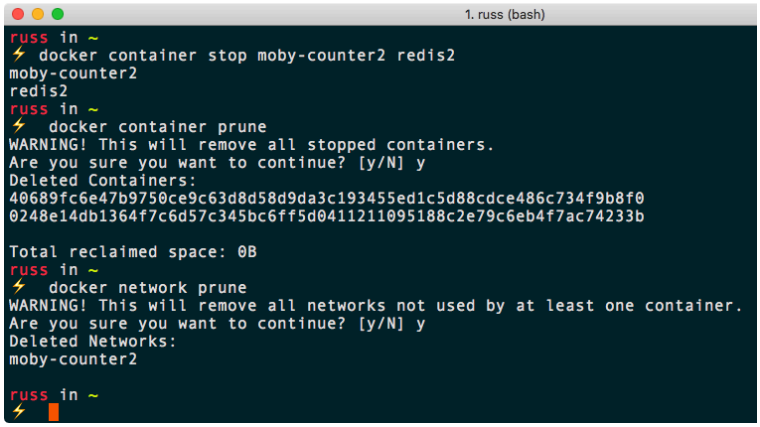
As you can see, it contains information on the network addressing being used in the IPAM section, along with details on each of the two containers running in the network.

IP address management (IPAM) *is a means of planning, tracking, and managing IP addresses within the network. IPAM has both DNS and DHCP services, so each service is notified of changes in the other. For example, DHCP assigns an address to* `container2`*. The DNS service is then updated to return the IP address assigned by DHCP whenever a lookup is made against* `container2`*.*

Before we progress to the next section, we should remove one of the applications and associated networks. To do this, run the following commands:

```
$ docker container stop moby-counter2 redis2
$ docker container prune
$ docker network prune
```

This will remove the containers and network, as shown in the following screenshot:



As mentioned at the start of this section, this is only the default network driver, meaning that we are restricted to our networks being available only on a single Docker host. In later chapters, we will look at how we can expand our Docker network across multiple hosts and even providers.